

Sovereign Debt Crisis

Module 2: The Eurozone Crisis

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Introduction

Three big questions

- **Trigger** — what turned trade imbalances into a crisis?
- **Propagation** — why did it spread and escalate?
- **Resolution** — what finally stopped it?

Two competing readings

- (i) Incomplete monetary union $\Rightarrow$ more institutional integration required
- (ii) Lack of discipline $\Rightarrow$ more rules and self-responsibility required

“Europe will be forged in crises, and will be the sum of the solutions adopted for those crises.”

— Jean Monnet

Introduction

Crisis narrative (Policy Insight 85)

- Build-up: current account imbalances, misallocated capital flows
- Triggers: US subprime + Greek deficit $\Rightarrow$ sudden stop within EMU
- Failed responses: austerity $\rightarrow \uparrow D/GDP \rightarrow \uparrow$ spreads
- Contagion: sovereign crisis + diabolic loop
- Denouement: ECB *whatever-it-takes* + banking union

Outline

2. A sudden stop — in a monetary union?
3. A (self-fulfilling) debt crisis?
4. The bank-sovereign *diabolic loop*

2. Sudden Stop in a Monetary Union

What is a sudden stop?

- Balance of payments identity: $CA + KA = 0$
 - ⇒ Any current account deficit must be financed by capital inflows
 - ⇒ BoP crisis: a country can no longer borrow to finance its CA deficit
- A **sudden stop**: sharp, rapid reversal of capital inflows
 - ↪ Usually triggers banking *and* currency crisis together
 - ↪ EM episodes: Mexico 1994, South-East Asia 1997, Russia 1998
- Self-validating dynamics:
 - Capital inflows → fuel growth and debt → attract more inflows
 - Reversal → triggers contraction → triggers further reversal

Refs: Merler & Pisani-Ferry (2012, Bruegel); Calvo, Izquierdo & Mejia (2004)

2. Sudden Stop in a Monetary Union

A common currency was supposed to make this impossible

- Sudden stops classically require a currency to break
- The canonical view — *One Market, One Money* (EC, 1990):

“A major effect of EMU is that balance-of-payments constraints will disappear (...) Private markets will finance all viable borrowers, and savings and investment balances will no longer be constraints at the national level.”

- ↪ Within a currency zone, nationality does not matter — only borrower solvency
- ↪ There is no Bavarian or Hamburger budget constraint in Germany
- ↪ The banking system and the CB smoothly allocate flows between regions
- ⇒ **Did it happen anyway?**

2. Sudden Stop in a Monetary Union

CA figures look smooth — but rewrite the BoP

$$CA + \underbrace{PKA}_{\text{private}} + \underbrace{T2 + PGM + SMP}_{\text{public}} = 0$$

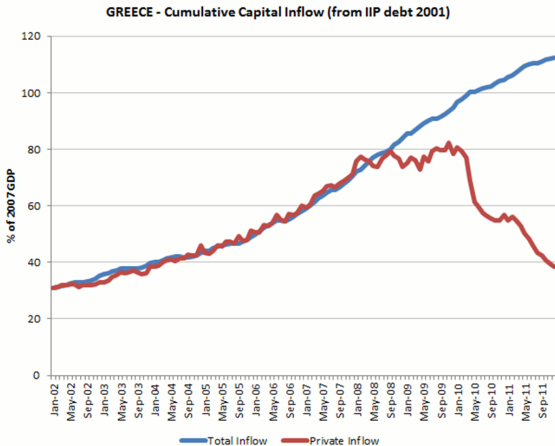
- *T2*: Target 2 inflows (ECB settlement system)
- *PGM*: EU/IMF assistance programmes
- *SMP*: ECB bond purchasing programme

Private capital fled; public flows silently replaced it.

The single currency did not prevent the sudden stop — it changed *who* absorbed it.

2. Sudden Stop in a Monetary Union

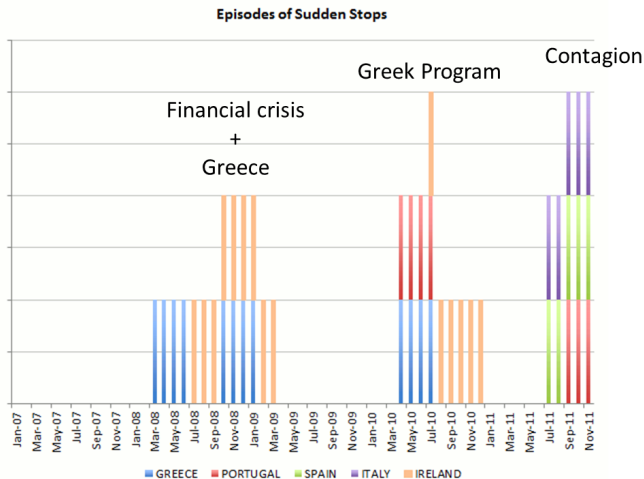
Greece: private outflows replaced by public inflows



Source: Merler & Pisani-Ferry (2012, Bruegel).

2. Sudden Stop in a Monetary Union

Cluster and contagion



Source: Merler & Pisani-Ferry (2012, Bruegel)

2. Sudden Stop in a Monetary Union

Three structural reasons

i. More a fixed exchange rate than an integrated area

↪ Redenomination risk made periphery assets effectively “foreign” for core investors

ii. The role of national banks and market segmentation

↪ Core banks reduced exposure to periphery risk — retreated home

↪ National supervision, national guarantees ⇒ national risk appetite

iii. Erratic political governance

↪ No credible backstop, no decisive institution

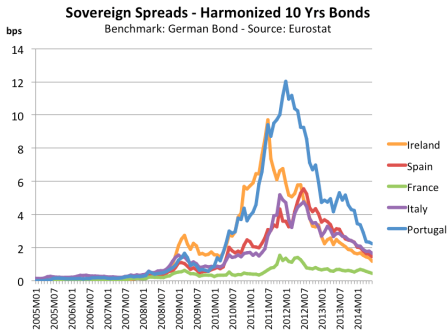
↪ Speculation fed on the uncertainty of “last-minute meetings of ultimate chance”

⇒ The sudden stop explains the *trigger*. But why did spreads keep rising long after? Could the crisis have been self-fulfilling?

Ref: Eichengreen & Wyplosz, *How the euro crisis was successfully resolved*, VoxEU

3. Self-Fulfilling Debt Crisis

Spread behaviour 2010–2012



... a common factor drives risk simultaneously across countries — difficult to explain by country-specific fundamentals alone ...

↳ What is a self-fulfilling debt crisis?

↳ Was it self-fulfilling in the Eurozone?

3. Self-Fulfilling Debt Crisis

Self-fulfilling when...



... economic outcomes are sensitive to *sentiment*, not only economic fundamentals ...

3. Self-Fulfilling Debt Crisis

A self-fulfilling debt crisis when...

- The price of government debt is sensitive to investors' sentiment:
 - ↪ If investors are optimistic: low interest rate $\Rightarrow$ no default ✓
 - ↪ If investors are pessimistic:
 - they charge a high interest rate ...
 - ... leading to default if the country is hit by a negative shock ...
 - ... which *justifies* the initial increase in the interest rate !
- $\Rightarrow$ No irrationality required to generate self-fulfilling crises.

Refs: Calvo (1988); Camous & Cooper

3. Self-Fulfilling Debt Crisis

Formally: a game between investors and the government

Consider a two-period economy:

- Today

- The government issues debt b
- Investors form beliefs about default $P(d)$ and charge rate $(1 + i)$:

$$(1 + i)(1 - P(d)) = R$$

- Tomorrow, given $(1 + i)b$ and income realisation $A \sim [A_l, A_h]$:

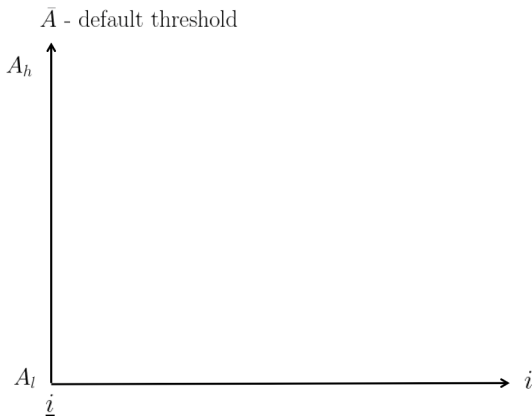
- Repay: raise labour taxes $\Rightarrow$ welfare $W^r(A, i, b)$
- Default (costly): welfare $W^d(A, i, b)$

$$\Delta(\cdot) = W^d(\cdot) - W^r(\cdot)$$

$\Rightarrow$ Default is optimal only when A is low (repayment cost is high)

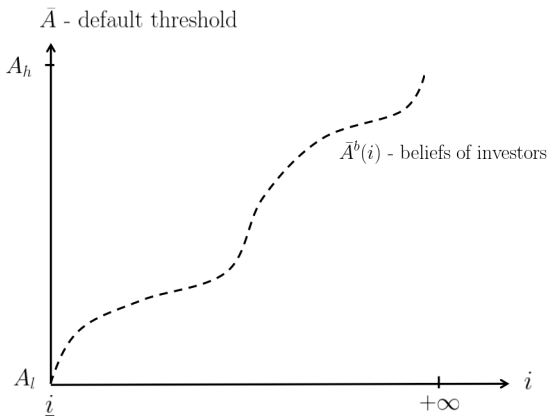
3. Self-Fulfilling Debt Crisis

Government defaults when the cost of repayment is too high



3. Self-Fulfilling Debt Crisis

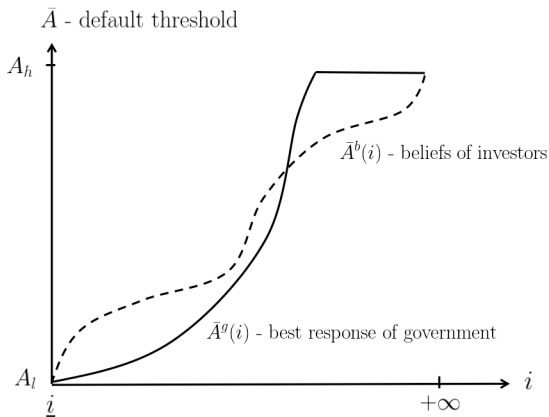
Investors form beliefs about $P(d)$ and charge $(1 + i)$



$$(1 + i)(1 - P(d)) = R$$

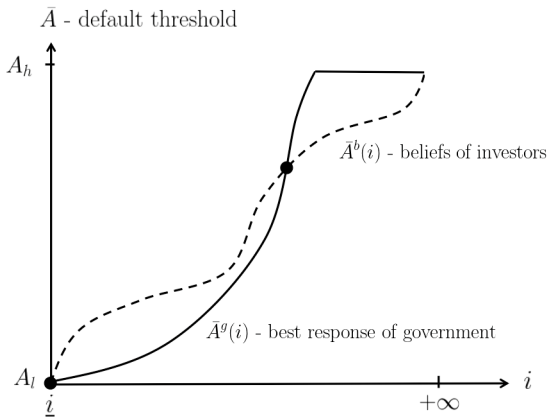
3. Self-Fulfilling Debt Crisis

In equilibrium, beliefs are rational: $P(d) = F(\bar{A})$



3. Self-Fulfilling Debt Crisis

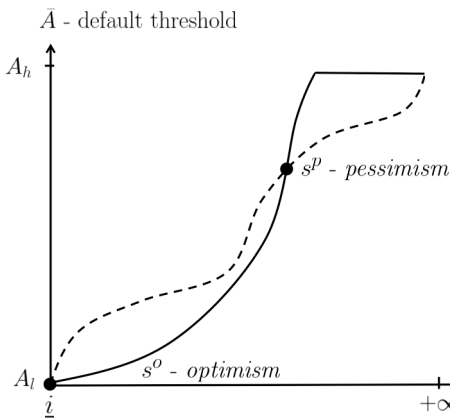
Multiple equilibria: good and bad



For an intermediate range of fundamentals, both a low-spread and a high-spread equilibrium are consistent with rational expectations.

3. Self-Fulfilling Debt Crisis

The sunspot: optimism or pessimism as a coordination device



Sunspot s captures investor sentiment. Which equilibrium prevails is determined by expectations alone — not by fundamentals.

3. Self-Fulfilling Debt Crisis

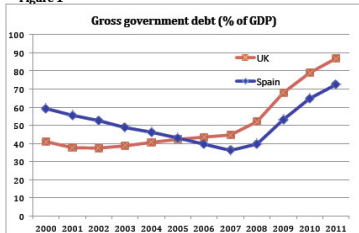
Was it self-fulfilling? — The policy stakes

Illiquid vs. *insolvent* — two very different diagnoses:

- Not self-fulfilling: spread rise reflects fundamentals $\Rightarrow$ need fiscal adjustment
- Self-fulfilling: need an institution to rule out the bad equilibrium
- Can be 50/50 — both implications hold simultaneously

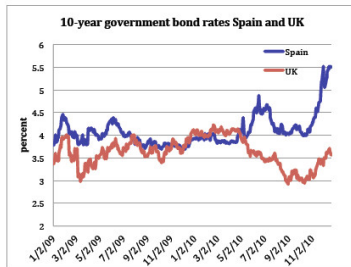
Why was Spain different from the UK?

Figure 1



Source: European Commission, Ameco

Figure 2:



Source: Datastream

Similar debt trajectories, banking crises, deep recessions — but the UK has a

3. Self-Fulfilling Debt Crisis

Testing for self-fulfilling dynamics: three obstacles

- i. We cannot directly observe *optimism* or *pessimism*
 - ii. The change in debt price *changes* fundamentals (endogeneity)
 - iii. Both fundamental and belief components can coexist
- ⇒ Can we disentangle changes in beliefs from changes in fundamentals?

Three complementary approaches

- (1) An **econometric** analysis — De Grauwe & Ji (2013)
- (2) An **accounting** exercise — Aldama (2021)
- (3) A **structural** analysis — Bocola & Dovis (2016)

3. Self-Fulfilling Debt Crisis

(1) Econometric: is there a spread component beyond fundamentals?

$$I_{it} = \alpha + z CA_{it} + \gamma Debt_{it} + \mu REE_{it} + \delta Growth_{it} + \alpha_i + \beta_t + \epsilon_{it}$$

- Fundamentals: current account, debt-to-GDP, real exchange rate, growth
- Country fixed effect α_i + time fixed effect β_t
- Estimated separately for EZ and standalone (non-EZ) countries

⇒ If fundamentals fully explain spreads, no EZ-specific component should appear

Ref: De Grauwe & Ji (2013)

3. Self-Fulfilling Debt Crisis

(1) Econometric results

Table 3

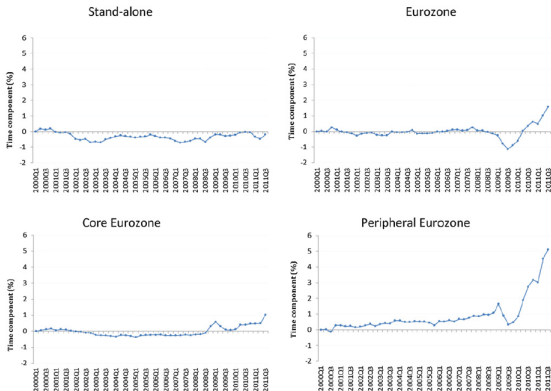
Spread and structural break in Eurozone.

	Pre-crisis	Post-crisis
Accumulated current account GDP ratio	0.0004 [0.0017]	-0.0809*** [0.0201]
Real effective exchange rate	-0.0135*** [0.0030]	0.2070** [0.0915]
Growth rate	-0.0005 [0.0037]	0.0053 [0.0266]
Debt/GDP ratio	0.0034 [0.0030]	0.1157*** [0.0248]
Country fixed effect	Controlled	Controlled
Observations	320	150
R^2	0.6758	0.8206

CA deficit and D/GDP ratios correlated with spread spikes post-2010 — the relationship is stronger and shifts specifically in the EZ. (Causality harder to assess...)

3. Self-Fulfilling Debt Crisis

(1) The time fixed effect β_t : a common EZ-specific component



"... this suggests that especially in the periphery, departures occurred in the spreads — an increase that cannot be accounted for by fundamental developments..."

⇒ Evidence of a sentiment or contagion component, distinct from

3. Self-Fulfilling Debt Crisis

(2) Accounting: decomposing the rise in D/GDP

Debt dynamics equation (from Module 1):

$$d_t - d_{t-1} = \frac{i_t}{1 + \gamma_t} d_{t-1} - \frac{\pi_t}{1 + \gamma_t} d_{t-1} - \frac{g_t}{1 + g_t} d_{t-1} + p_t$$

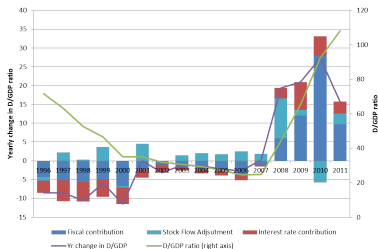
- Interest rate i_t — potentially self-fulfilling channel
- Inflation π_t and real growth g_t
- Primary deficit p_t

⇒ How much of the rise in D/GDP is explained by the interest rate channel?
Compare Ireland (bank bailout origin) vs. Greece (fiscal origin)

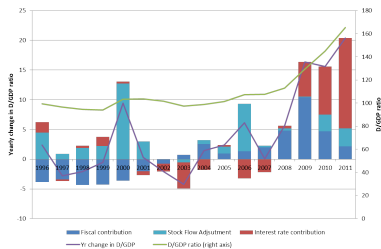
Refs: IMF (2010); Aldama, OFCE (2021)

3. Self-Fulfilling Debt Crisis

(2) Accounting: Ireland and Greece



Ireland: SFA (bank bailout)
dominates



Greece: interest cost + primary
balance

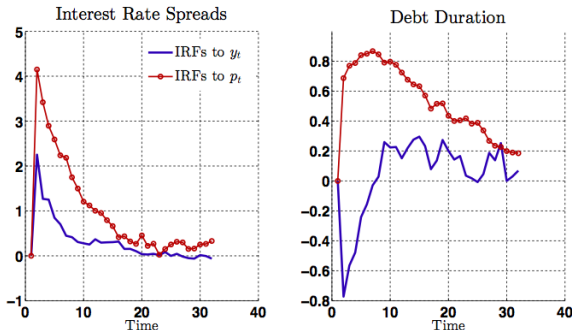
In Greece, the interest rate channel appears non-negligible — one possible reading is that a self-fulfilling component may have contributed to the debt dynamics, though disentangling it from fundamentals remains difficult.

3. Self-Fulfilling Debt Crisis

(3) Structural identification — Bocola & Dovis (2016)

- Two key problems for identification:
 - Endogeneity of prices and outcomes
 - Sentiments are not observable
- The key insight: **debt maturity as an identification device**

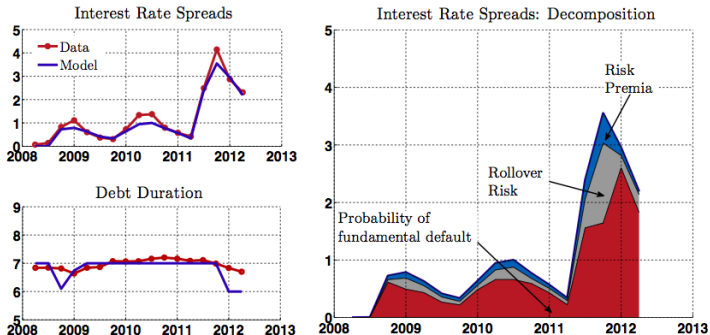
Figure 1: The dynamics of interest rate spreads and debt duration



3. Self-Fulfilling Debt Crisis

(3) Application to Italy 2008–2012

Figure 5: Contribution of rollover risk to interest rate spreads



- LHS: model fit to observed spread and maturity data
- RHS: decomposition — a significant share of the 2011–12 spread spike is attributed to the *belief component*, especially at the peak

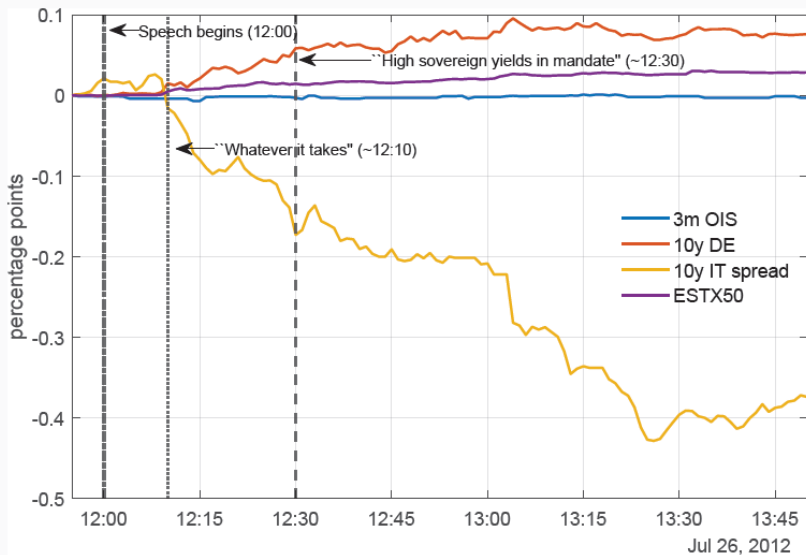
3. Self-Fulfilling Debt Crisis

Resolution: ruling out the bad equilibrium

- Spring 2012: coordinated government interventions keep failing
- July 2012 — Mario Draghi:

“Within our mandates, the ECB is ready to do whatever it takes to preserve the euro. And believe me, it will be enough.”

- ECB commits to unlimited sovereign bond purchases (OMT)
 - Never needs to implement it $\Rightarrow$ an effective commitment device
- $\Rightarrow$ Theory predicts exactly this: a credible backstop eliminates the bad equilibrium without requiring actual purchases



4. The Diabolic Loop

“We must break the vicious cycle of banks hurting sovereigns and sovereigns hurting banks. This works both ways.”

— Christine Lagarde, April 2012

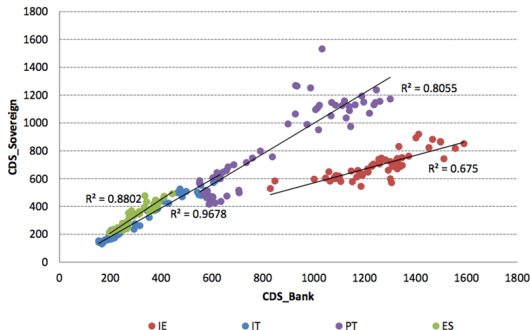
“Rarely does an economic idea gather so wide a consensus as the evilness of the “deadly embrace”. The feedback loop between weak bank balance sheets and sovereign fragility now faces almost universal opprobrium (...) providing a major impetus to build the European banking union.”

— Farhi & Tirole (2014)

4. The Diabolic Loop

Bank–sovereign credit risk: correlation during the crisis

Correlation of bank and sovereign CDS — Jan 2011 to Feb 2012



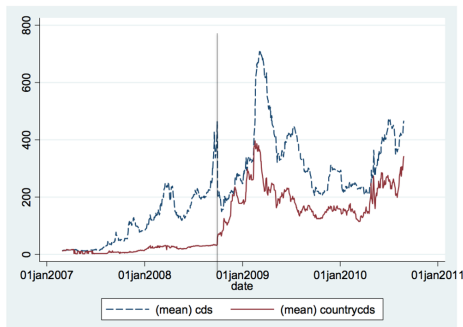
Correlation was near zero before 2011. During the crisis: strongly positive, both directions.

⇒ **Causality runs both ways.**

4. The Diabolic Loop

Channel 1: banks hurting sovereigns (bailout channel)

Ireland — September 2008

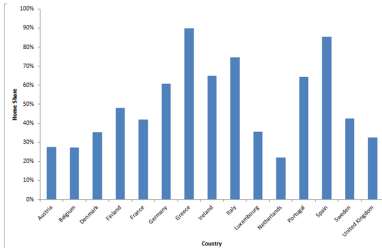


Acharya, Drechsler & Schnabl (VoxEU, 2012)

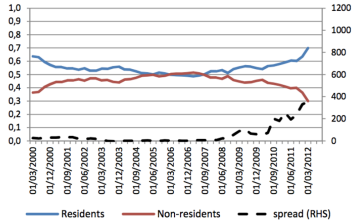
⇒ Bank credit risk ↑ → implied probability of government bailout ↑ → sovereign credit risk ↑

4. The Diabolic Loop

Channel 2: sovereigns hurting banks (domestic holdings)



Home bias in sovereign bond holdings (March 2010). Banks held disproportionate shares of domestic debt.
Acharya et al. (2012)



Spain: domestic holding of sovereign debt *increased* during the crisis.

Broner, Erce, Martin & Ventura (VoxEU, 2014)

⇒ Sovereign impairment directly damages bank balance sheets — and exposure *rose* as risk rose. Why?

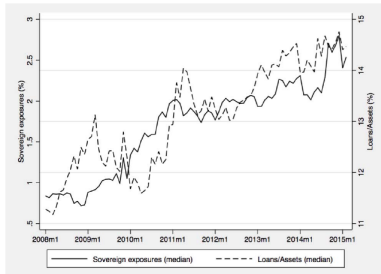
4. The Diabolic Loop

Why do banks hold (and accumulate) domestic sovereign debt?

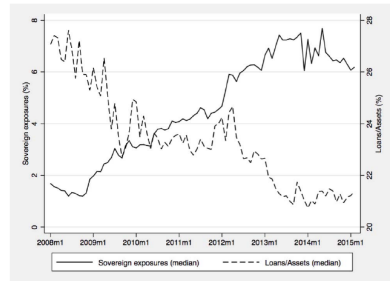
- i. Financial repression: sovereigns use cheap ECB liquidity to fund bond purchases
 - ↪ *“This [LTRO] means that each state can turn to its banks, which will have liquidity to buy the bonds” (Sarkozy)*
- ii. Financial dominance: banks accumulate domestic bonds to guarantee bailout
 - ↪ “Strength through weakness” — too systemically exposed to be allowed to fail
- iii. Creditor discrimination: domestic holders expect preferential treatment in default
 - ↪ Spain would compensate domestic banks to limit contagion costs
- iv. Risk shifting: concentrate risk domestically to arm-twist an ECB/European bailout

4. The Diabolic Loop

Transmission to the real economy: credit crowding-out



Non-stressed countries



Stressed countries (PT, IR, GR, IT, ES)

Domestic holding of sovereign debt vs. loans to non-financial corporations.

In stressed countries: rising sovereign exposure crowds out bank lending to the private sector.

⇒ The diabolic loop is not merely financial — it produced the real recession.

Source: ECB Working Paper 1969 — Bank exposures and sovereign stress transmission

4. The Diabolic Loop

Resolution: breaking the loop through banking union

- The loop exists because banks and sovereigns are *national*:
 - National supervision, national guarantees, national bailout
 - Sovereign distress $\Leftrightarrow$ bank distress
 - **Banking union**: Europeanise the banking system
 - **SSM**: common supervisor (ECB)
 - **SRM**: common resolution mechanism — from bail-out to bail-in
 - **DGS**: harmonised deposit guarantees
 - From national banks to European banks
- ⇒ Banking union is the structural institutional response to the missing integration that made the loop possible

Key Takeaways

- ① A **sudden stop** did occur inside the euro area — the single currency changed its *form* (public flows absorbed what private flows fled) but not its *nature*.
- ② **Self-fulfilling dynamics** were quantitatively significant, especially in 2011–12. The EZ was special: no lender of last resort for sovereigns made the bad equilibrium viable.
- ③ **OMT worked** because it changed the game — not by purchasing bonds, but by making the bad equilibrium non-credible. Theory predicts exactly this.
- ④ The **diabolic loop** shows why incomplete monetary union amplified every shock: national banks and national guarantees made sovereign and bank risk inseparable.
- ⑤ **Banking union** is the structural institutional fix — the response to the same missing integration that made the loop possible.

References

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